

Cedric Honnet

honnet@mit.edu // www.honnet.eu

ABSTRACT

With endless enthusiasm about interactivity and miniature wearables, I love to explore new technologies... I am deeply curious and always open for collaborations.

Keywords

HCI, Miniature Embedded Systems, Manufacturing, eTextiles, 3d [Position | Motion | Touch] Sensing, [Ultra] Low Power, Wearables, Haptics, Arts & Science, Human Augmentation...

RESEARCH

Selected Publications (more details: honnet.eu/#Academic)

- UIST 2025: *FiberCircuits: A Miniaturization Framework To Manufacture Fibers That Embed Integrated Circuits*
- UIST 2024: *PortaChrome: A Portable Contact Light Source for Integrated Re-Programmable Multi-Color Textures*
- UIST 2023: *MagKnitic: Machine-knitted Passive and Interactive Haptics Textiles with Integrated Binary Sensing*
- IEEE ICRA 2023: *Laser-Etching Flexible Sensors for Robotic Touch Recognition*
- UIST 2023: *FibeRobo: Fabricating 4D Fiber Interfaces by Continuous Drawing of Temperature Tunable Liquid Crystal Elastomers*
- AHs 2022: *A Framework For Designing Augmented Human-Flora Interaction*
- UIST 2021: *MetaSense: Integrating Sensing Capabilities into Mechanical Metamaterial*
- IEEE AeroSpace 2021: *Feeling Through Spacesuits: Application of Space-Resilient E-Textiles to Enable Haptic Feedback on Pressurized Extravehicular Suits*
- CHI 2020: *PolySense: Augmenting Textiles with Electrical Functionality using In-Situ Polymerization*
- UIST 2020: *Sonoflex: Embroidered Speakers Without Permanent Magnets*
- TEI 2019: *Optimizing Pressure Matrices: Interdigitation and Interpolation Methods for Continuous Position Input*
- AH 2018: *HIVE Tracker: a tiny, low-cost, and scalable device for sub-millimetric 3D positioning*
- NIME 2017: *Designing a Multi-Touch eTextile for Music Performances*
- LM 2016: *Developing an Ecosystem for Interactive Electronic Implants*

EXPERIENCE

2022: Samsung Research, USA

Human-Computer Interactions & Embedded systems, Intern Think Tank Team: Touch Sensing, Haptics, Wearables...

2019-2020: MIT - NeuroGEARS

Human-Computer Interactions, Embedded systems
Deployed as visiting scientist in the Media Lab.
3D positioning, Wearables...

2017-2019: Sorbonne University / CNRS, France

Research Associate/Affiliate - Embedded systems
"Situated Analysis of HCIs in Smart Environments"

2016-2019: Carpe Noctem, International

Research Engineering, Mentor
- Consulting about embedded systems, wearables, IOT and manufacturing in China, France, etc.
- Visiting Researcher in UK, Denmark, Singapore, Portugal
- Mentoring/teaching in grad schools, incubators...

2014-2016: Tangible Display, USA / France / China

Co-founder - Research & Development Engineer
- Helped companies exploring various interactive solutions for their project (ex: Google Soli)
- Designed interactive systems (hard+soft) with RFID, touch screens, depth cameras, wearables, sensor fusion, BLE, etc

2012-2013: Sifteo, USA

Embedded Systems Engineer & InterHacktivist
- Firmware and software: user interface (sound, graphics...), battery management, simulator, graphics controller...
- Experimental interactive projects: gestures recognition, music control, audio communication system...

2010-2011: Sigma Designs - USA

Research & Development Engineer, Trainee
Google's VP8 Video Decoder Porting on Custom Processor:
- Firmware: memory management and execution acceleration
- Hardware instruction customization for specific purposes such as arithmetic decoding or binary tree decoding

2008: Telecom ParisTech EE Research Labs - France

Research & Development Engineer, Trainee
Evaluation of combinatorial components reliability using FPGA for massively parallel simulation of fault injection.

2007: Brunel University EE Research Labs - UK

Research & Development Engineer, Trainee
Signal Restoration experimentations & Universal Software Radio Peripheral modules implementations.

EDUCATION

2021-2027: MIT, USA

PhD Program @ Media Lab + CSAIL

(Master: fast tracked - GPA: 5.0/5.0)

Tangible Media + HCI Eng + Responsive Environments

Extracurricular: Kick Boxing + Brazilian jiu-jitsu

2011: Telecom ParisTech (IPP), France

★ 1st French Engineering school ([Industry ranking](#))

Master 2 - with honors (Diplome d'ingenieur)

Computer Sciences: Embedded Systems & Robotics, New Media, Systems On Chip, Security

Extracurricular: Robotics, Savate (French boxing)

2009: Sorbonne University (ex-UPMC), France

★ 1st French university ([Research ranking](#))

Master 1 - with honors (Maitrise)

EECS with emphasis on Digital Electronics

Extracurricular: Jujitsu, Sambo, Computer Science tutoring for students with disabilities

2007: Brunel University, UK

Bachelor - with honors

EECS with emphasis on Digital Signal Processing

Extracurricular: Judo, Thai Boxing

LANGUAGES

- French: ILR Level 5 - Native proficiency
- English: ILR Level 4 - Full professional proficiency
- Spanish: ILR Level 3 - Professional working proficiency

DISTINCTIONS, AWARDS...

- 2025: [UbiComp/ISWC](#) award for [FerroZuit](#)
- 2025: [Amazon Fellowship](#) for [FiberCircuits](#) Next Gen.
- 2024: MIT [Fast Company](#) Innovation Award
- 2023: MIT [MPP](#) grant for photovoltaic textiles
- 2022: MIT ML [Nagashima](#) Fellowship
- 2021: MetaSense - [Wildcard Prize](#) by Hack-a-Day (HaD)
- 2020: PolySense - [Wildcard Prize](#) by HaD
- 2020: Selected for the [Open Lab](#) with Google Arts
- 2019: [NYU-ITP](#) "instigator" fellowship
- 2019: HiveTracker - [Connected World](#): Best Project (HaD)
- 2018: HiveTracker - [HCI](#) Prize (HaD)
- 2017: French government research grant ([CIR](#))
- 2016: Mentor in art+tech accelerator ([MaDlab](#) UK.)
- 2015: [Google Soli](#) alpha development fellow
- 2015: [Forbes](#) featured my project Twiz!
- 2014: 1st place at Bemyapp factory [contest](#), SF, CA
- 2009: [Besse Foundation](#) Fellow (until 2011)
- 2005: Kenpoken France champion - [80 Kg category](#)

EXTRA

Academic Impact Amplification

- *Yearly Research Residency* - [Research At Scale](#) (organizer): Dozens of MIT researchers spending a month in Asia to scale their work, and explore manufacturing ecosystems...
- *Yearly Symposium* - [ScalableHCI](#) (co-founder + organizer): Hundreds of international participants experiencing a compressed version of the Research At Scale program in 1 week.

[Meta]Reviewing

- UIST: User Interface Systems and Technologies
- UbiComp/ISWC: Ubiquitous and Wearable Computing
- CHI: Human Factors in Computing Systems
- TEI: Tangible Embedded and Embodied Interaction
- NIME: New Interfaces for Musical Expression

Collectives

- [DataPaLETTE](#): wearables/e-textile hackerspace (founder)
- [Fuz](#): hackerspace against planned obsolescence (founder)
- [Noisebridge](#) hackerspace: teaching electronics
- [Cumbia Rockers Allstars](#) ([Gatos Negros Prod](#)): DJ in Cumbia-Fusion groups
- [Telecom Robotics](#): inter grad school competitions

Art-Science Residencies & Exhibitions

- Austria 2020: ARS Electronica - [Topographie Digitale](#) exhibition
- US - MA 2018: MIT Media Lab research residency about wearables & music (gave birth to [SonoFlex](#))
- Singapore 2016: Media Lab Lasalle - visiting researcher: see [Digital Oxymorons](#) publication.
- Denmark 2015: Copenhagen University, Human-Centred Computing - visiting researcher: [body-ui.eu](#)
- France 2015: Fructus Animalis - art against programmed obsolescence : [project.honnet.eu/fructus-animalis](#)
- Singapore 2014: Growell Hackerspace - explored wild animals camera-trap, DIY farm robotics...
- Colombia 2012: [Plataforma - Bogota](#) (Interactive Art Lab): [project.honnet.eu/Resbala](#), etc.

REFERENCES

- Joe Paradiso (MIT): [JoeP\(a\)media.mit.edu](mailto:JoeP(a)media.mit.edu)
- David Merrill (Sifteo): [dMerrill\(a\)media.mit.edu](mailto:dMerrill(a)media.mit.edu)
- Ali Rizvi (Sigma Designs): [aRizvi\(a\)apple.com](mailto:aRizvi(a)apple.com)
- Alexis Polti: [Alexis.Polti\(a\)Telecom-ParisTech.fr](mailto:Alexis.Polti(a)Telecom-ParisTech.fr)
- Saeed Vaseghi: [Saeed.Vaseghi\(a\)Brunel.ac.uk](mailto:Saeed.Vaseghi(a)Brunel.ac.uk)